

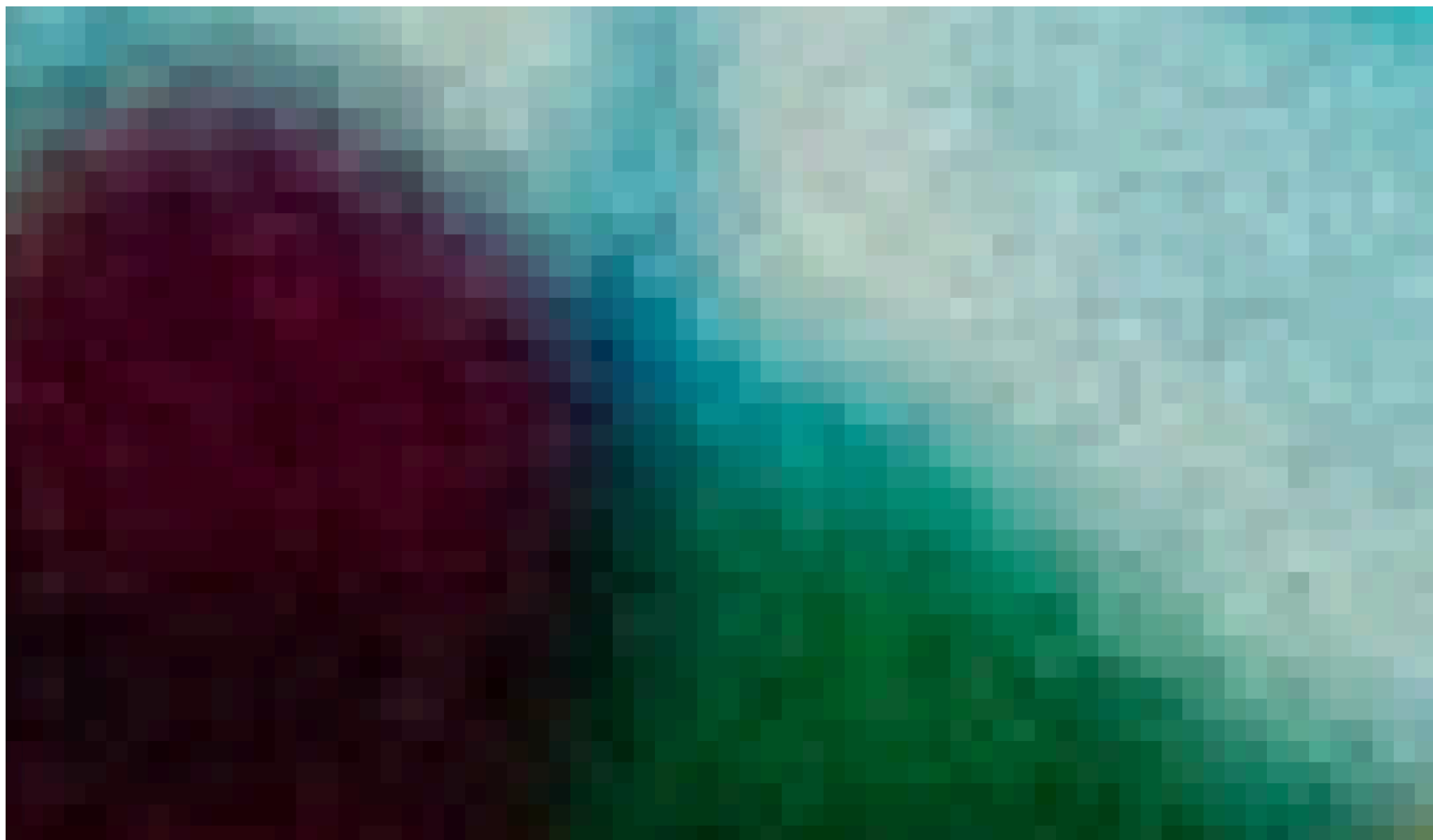
Geographic Data Science

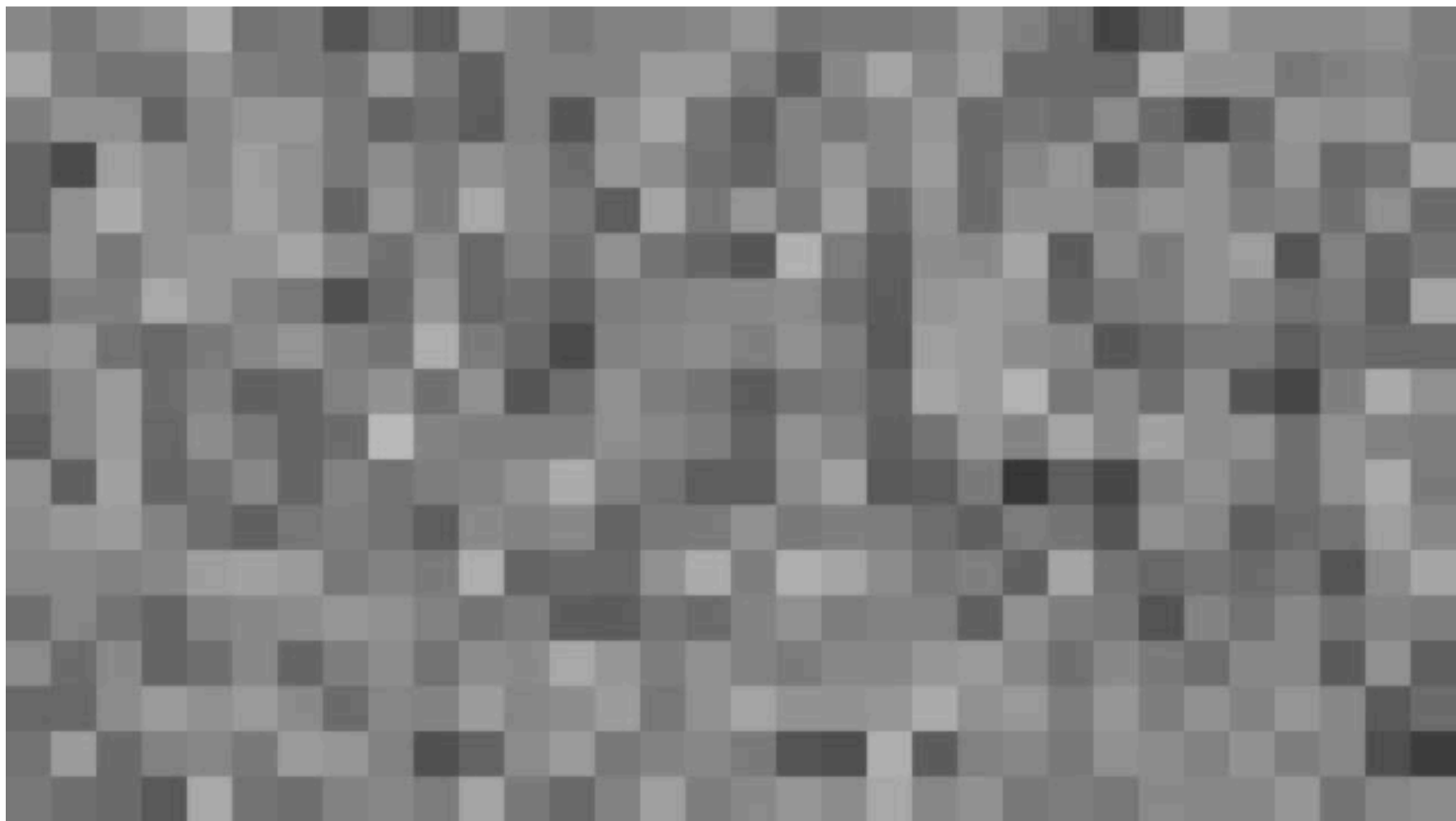
Raster Data

Elisabetta Pietrostefani & Carmen Cabrera

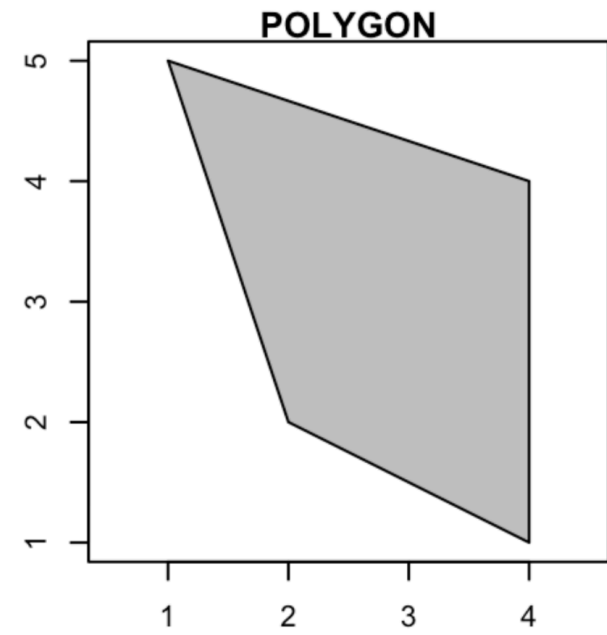
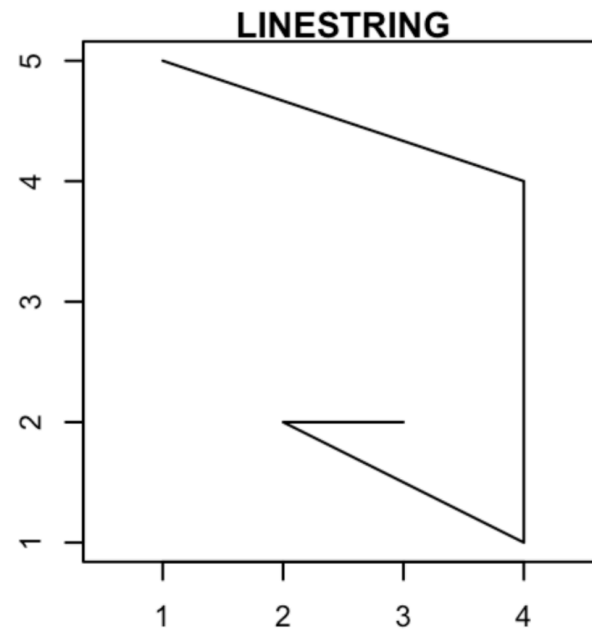
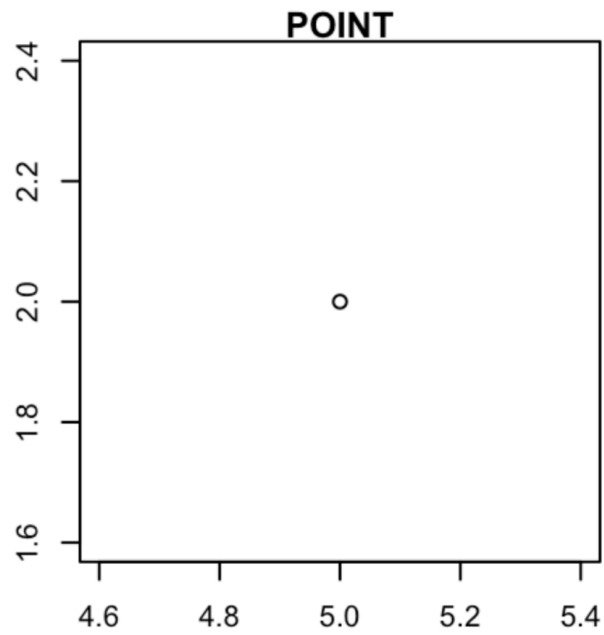
Raster Data



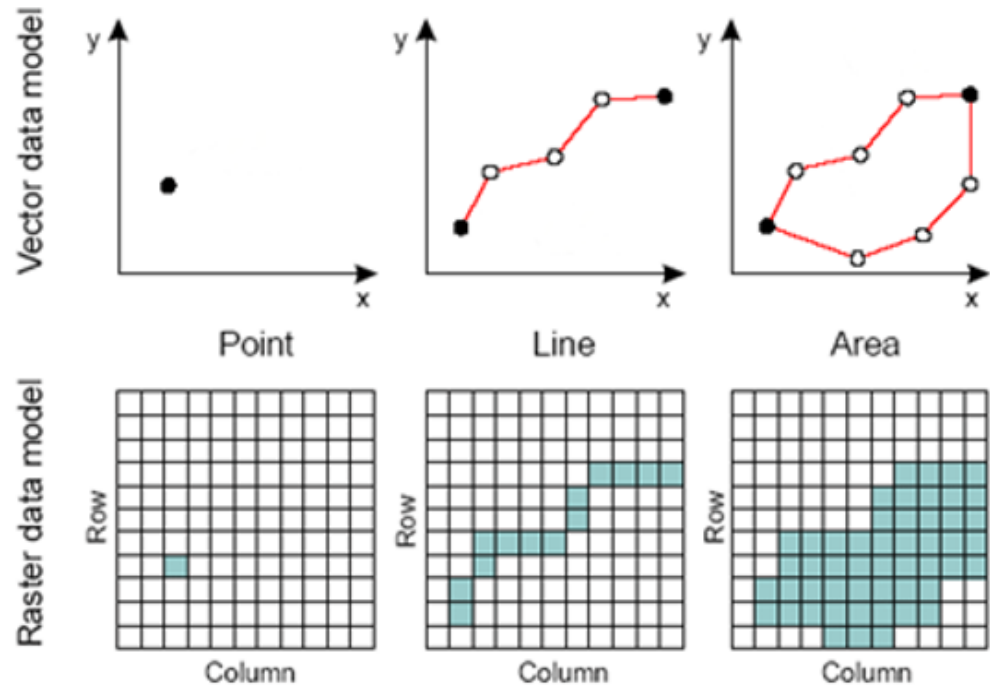




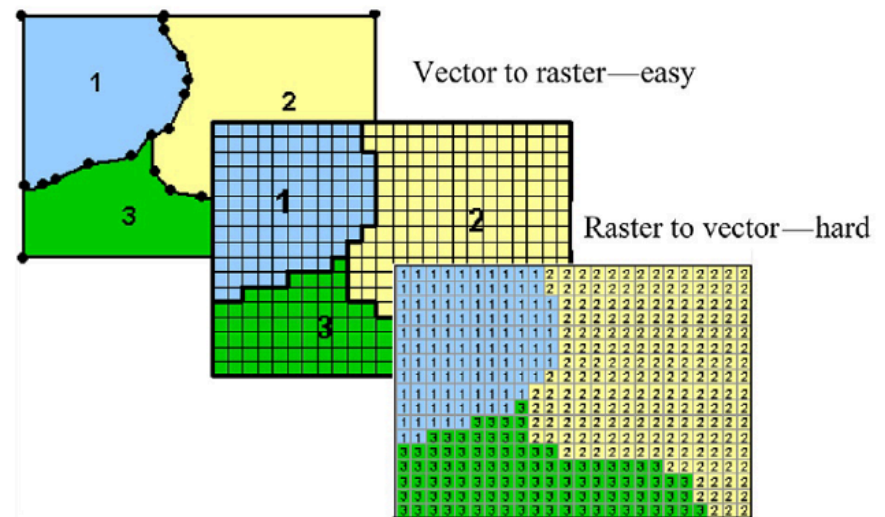
Vector Data



Contrast with Vector Data



Most GIS software permit raster-vector format conversions:

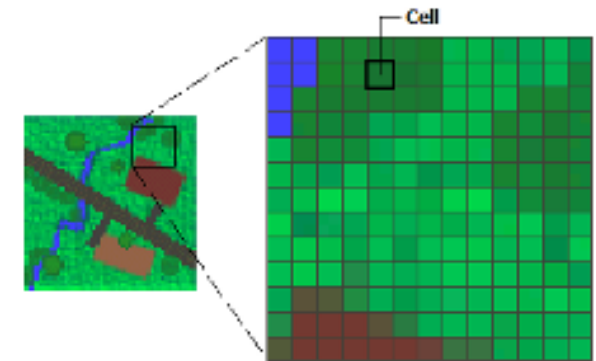
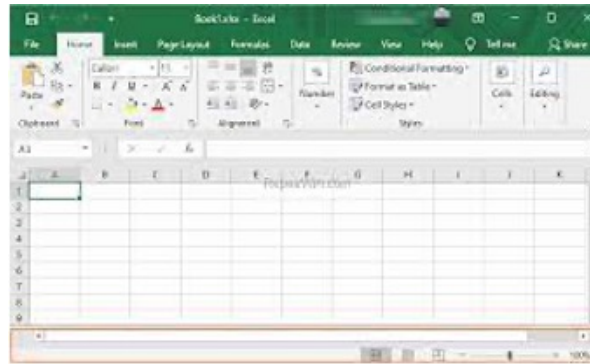


Source: FAO

Definition

- Square grid of pixels. Pixel values can represent continuous or categorical variables:
- Divides 2-D space into regular cells - pixels
- Each cell has a single value
- Values assigned according to value at mean, centre point, or some other rule

File Formats



Raster Data Types

Grayscale Rasters

Continuous Data, Single Band



Nightlights data

- Nightlights data can be represented as a grayscale raster, where darker areas indicate lower levels of artificial light, and lighter areas represent higher levels of artificial light.
- The pixel values may represent the radiance or luminance values of nighttime lights.
- Used for monitoring urban development, assessing light pollution, and understanding human activity patterns at night...

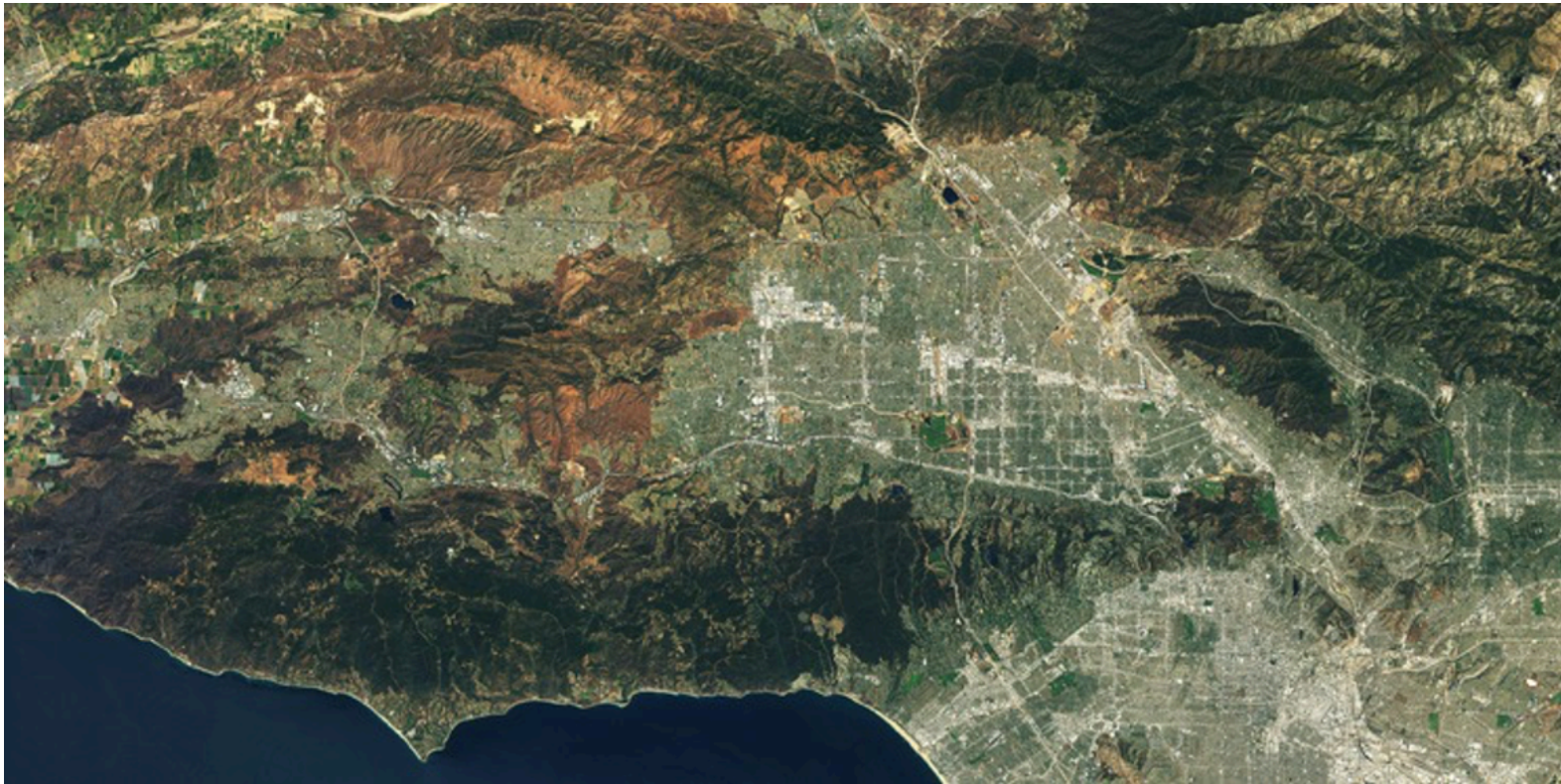
Multispectral Rasters

Multiple Bands

Band Number	μm	Resolution
1	0.433–0.453	30 m
2	0.450–0.515	30 m
3	0.525–0.600	30 m
4	0.630–0.680	30 m
5	0.845–0.885	30 m
6	1.560–1.660	30 m
7	2.100–2.300	30 m
8	0.500–0.680	15 m
9	1.360–1.390	30 m
10	10.6–11.2	100 m
11	11.5–12.5	100 m

Bands 2, 3 and 4

Blue, green, and red for true-color image.



Landsat Satellite Imagery

- Landsat satellites capture data in multiple spectral bands, such as visible, near-infrared, and thermal infrared.
- Each band is represented as a separate channel in a multispectral raster, allowing for the analysis of various aspects of the Earth's surface.
- Different bands capture information related to vegetation, water, and land use.

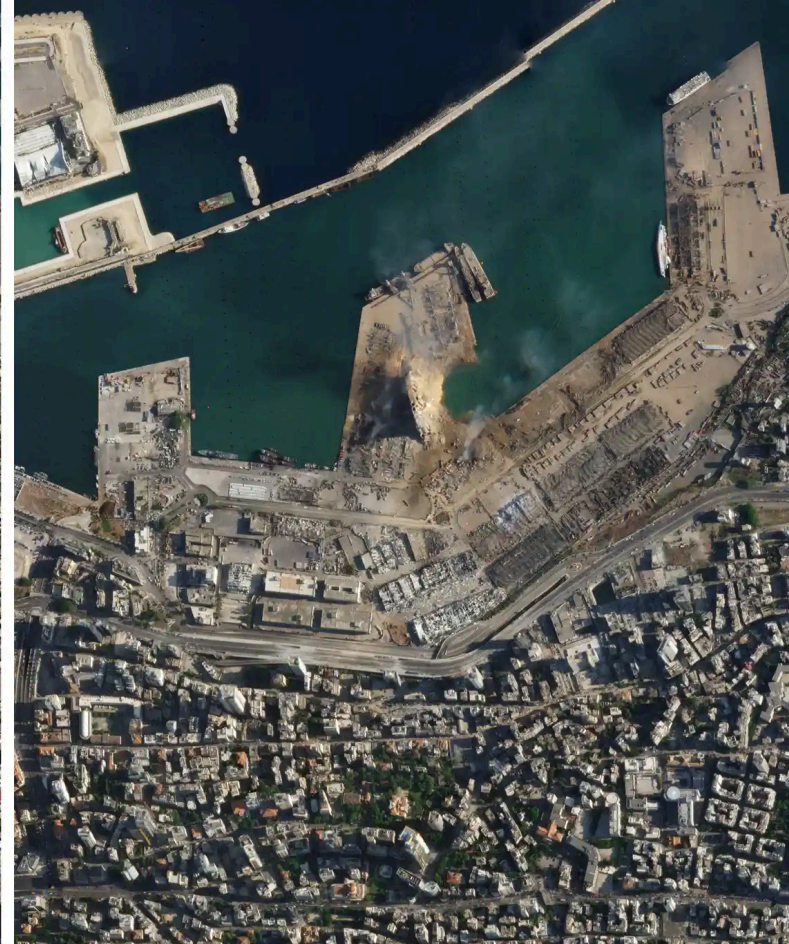
Sources

- High spatial resolution data - costly commercial products.
- Lower spatial resolution data is free (NASA, ESA, etc).
- Several sources:
 - [USGS Earth Explorer](#) — Landsat, and much else
 - [NASA Earthdata Search](#)
 - [AppEEARS](#) — extract by area, no bulk download
 - [LP DAAC Data Pool](#)
 - [Copernicus Data Space](#) — Sentinel (replaced the old SciHub)
 - [Landsat on AWS](#)

Several of these need a free account, and all involve downloading large files.

Color Rasters

Digital Photography, Optical Satellites

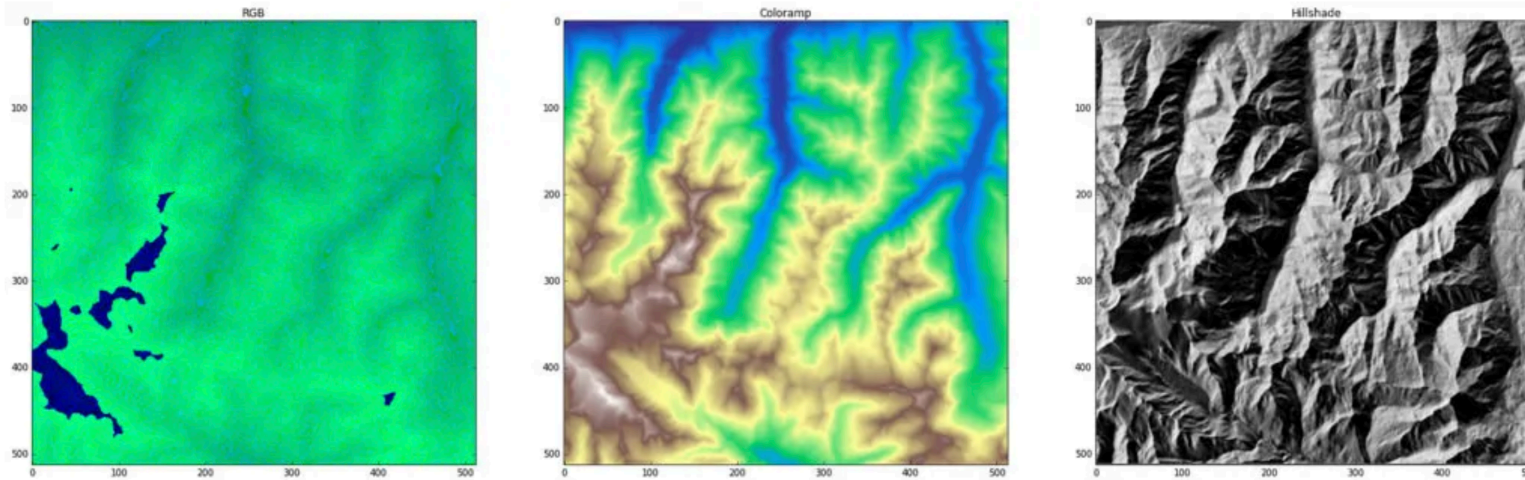


Color Rasters

- Digital photographs are typically stored as color rasters, combining red, green, and blue (RGB) channels to represent a full range of colors.
- Each pixel's RGB values determine the color in the image.
- Digital cameras and most image-editing software work with RGB color rasters.

Elevation Rasters

Representing Terrain Data



Left: *raw RGB*, Center: *decoded to hypsometric tint*, Right: *decoded to hillshades*

Mapbox

Digital Elevation Models (DEMs)

- Digital Elevation Models are used to represent terrain data.
- DEMs are rasters where each pixel's value represents the elevation (height) of the corresponding location on the Earth's surface.
- These rasters are widely used in geospatial applications, including topographic mapping and hydrological modeling.

Resolution

Spatial Resolution

- Level of detail or granularity in the spatial domain of data.
- How well an instrument or sensor can distinguish between objects or features in a given space.
- Higher spatial resolution = more detailed data
- Larger file sizes and Increased data processing demands.

Temporal Resolution

- Accuracy and precision of time measurements
- The higher the temporal resolution, the more frequent and precise the time measurements, allowing for the tracking of fast-changing processes or events.

Data Volume and Storage

- Unprecedented rate of collection.
- Fundamental aspects of data management.
- Challenges associated with handling and storing vast amounts of data
- Traditional hardware-based solutions vs cloud-based storage (right approach?)

Future Trends

Big Data and Raster Information

- Transforming the way we handle geospatial data.
- Sensors and technology become more sophisticated
- The volume of raster information is growing exponentially.

Machine Learning and Raster Data

- Chen Wangyang, Wu, Abraham Noah, Biljecki and Filip, 2021. Classification of Urban Morphology with Deep Learning: Application on Urban Vitality. *Computers, Environment and Urban Systems*

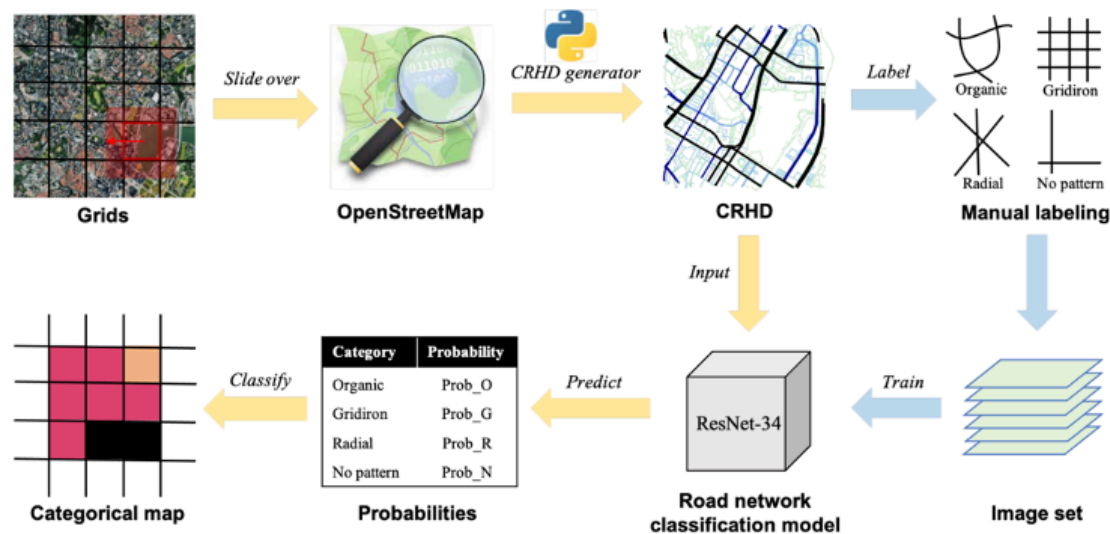


Figure 3: Process of road network classification.

Cloud-Based Raster Data Services

- Redefining how we store, process, and share raster data.
- The cloud offers scalability, accessibility, and cost-efficiency that traditional on-premises systems can't match.

Satellite data for Social Science

Henderson et al 2012.

“Measuring economic growth from outer space”, AER.

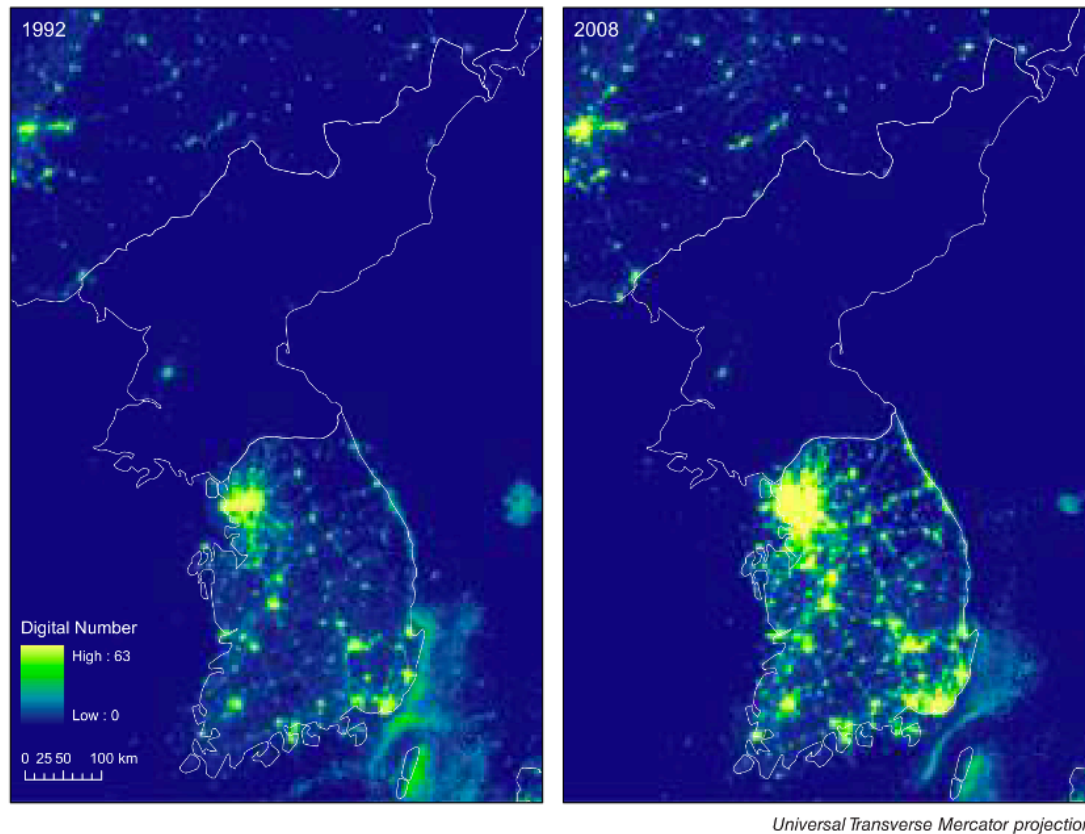
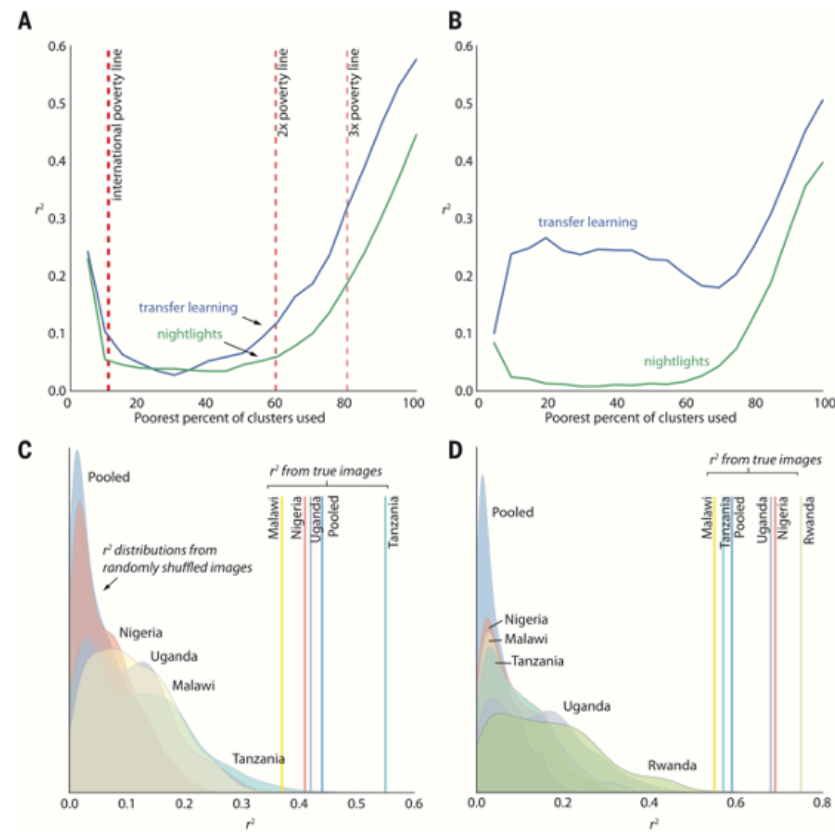


FIGURE 2. LONG-TERM GROWTH: KOREAN PENINSULA

Jean et al. 2016.

“Combining satellite imagery and machine learning to predict poverty.” Science



From pixels to policy

The problem with raw imagery

Everything on the last few slides sounds great — but working with raw satellite data is *hard*

- A single Sentinel-2 scene is **several gigabytes**
- Half your pixels are hidden under **cloud**
- Cleaning, mosaicking and modelling needs real compute
- And pixels don't line up with the units social science actually uses — LSOAs, wards, local authorities

The four steps

Turning a noisy grid of numbers into something usable:

1. **Raw** — spectral values, noisy, partly cloud-covered
2. **Clean** — combine many passes to remove cloud
3. **Model** — convert spectral values into a meaningful indicator (temperature, vegetation, cloud probability)
4. **Aggregate** — pool pixels into administrative units

Step 4 is where raster becomes choropleth

This is what Imago does

Imago is the imagery data service for sustainability, prosperity and wellbeing — part of Smart Data Research UK, based partly here at Liverpool

They do steps 1–4 for you and publish the result:

- Ready-to-use **LSOA/MSOA-level** statistics
- No gigabyte downloads, no remote-sensing algorithms
- Small-area detail preserved, not regional averages

You've already used this

The **SPF (Sun Probability Framework)** data you mapped in the choropleths lab?

That's satellite-derived cloud probability, aggregated to small areas.

You were doing raster analysis without touching a raster

Which is the point — someone still has to do the pixel work. This week, that's you.

Go deeper

Imago training — free, open, in R and Python.
Air temperature, precipitation, SPF. Several
run in your browser, no install.

imago-sdruk.github.io/Imago_training



From grids to areas

The full walkthrough of everything on the last few slides — spectral bands, resolution, revisit cycles, and the raw → clean → model → aggregate pipeline.

[Imago: From grids to areas](#)



In the lab

What you'll do

Terrain — Lebanon

- Load a DEM, check its CRS, reproject it
- **Crop** then **mask** to the country boundary
- Style it properly (breaks, elevation palettes)
- **Extract** raster values to survey points — the raster→vector bridge

Night lights — Kenya & Tanzania

- Stack multi-year rasters
- **Zonal statistics:** mean light per administrative region
- Map 1992 vs 2013 — and find the story in it

Two things to watch for

1. **CRS mismatches.** The single most common reason raster code silently returns **NA**. Check before you extract, not after.
2. **Crop *then* mask.** Both narrow a raster to your area of interest, but cropping first is much faster on large files.

Questions



Geographic Data Science by [Elisabetta Pietrostefani](#) is licensed under a [Creative Commons Attribution-ShareAlike 4.0 International License](#).